A feedback loop with a communication delay

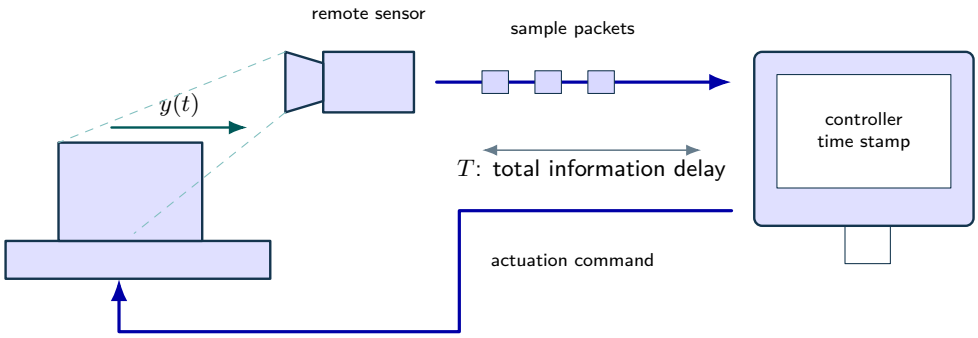
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

The normalized tracking loop is $L(s) = 8/[s(s+3)]$. Find gain crossover ω_c, phase margin ϕ_m and the first constant pure-delay stability boundary. Introduce delays $T = 0.10\,\mathrm{s}$ and $0.30\,\mathrm{s}$. Compare fifth-order Pade approximation with exact phase $-\omega T$, explaining the scope of the rational approximation.

2 Physical system



Mechanism and communication path; packet losses and jitter are omitted

A remote mechanism sends measurements and receives commands through a communication path. The model combines those latencies into one ideal constant delay; the drawn packet train is not an experimental record.

3 Gain crossover

The open-loop magnitude is $8/[\omega\sqrt{\omega^2+9}]$. Setting it to one and defining $z = \omega^2$ gives $z^2 + 9z - 64 = 0$, hence

$$\omega_c = \sqrt{\frac{-9+\sqrt{337}}{2}}.$$

The phase is $-90^\circ - \tan^{-1}(\omega/3)$, so the no-delay phase margin is $90^\circ - \tan^{-1}(\omega_c/3)$.

4 Delay consumes phase

A pure delay multiplies the loop by e^{-sT}. Its frequency magnitude is one and its phase is $-\omega T$ radians. Therefore crossover does not move and

$$PM(T) = PM(0) - \omega_c T\frac{180}{\pi}, \qquad T_{crit} = \frac{PM(0)\pi/180}{\omega_c}.$$

This is the first imaginary-axis crossing for this stable nominal closed loop with its single positive gain crossover. It is not a universal rule for arbitrary loops with multiple crossings.

5 Rational approximation

The fifth-order Pade model approximates the delay by a rational all-pass transfer function. Compare its phase at crossover with the exact phase before using its time response. Artificial right-half-plane zeros and high-order dynamics are approximation features, not physical actuator modes.

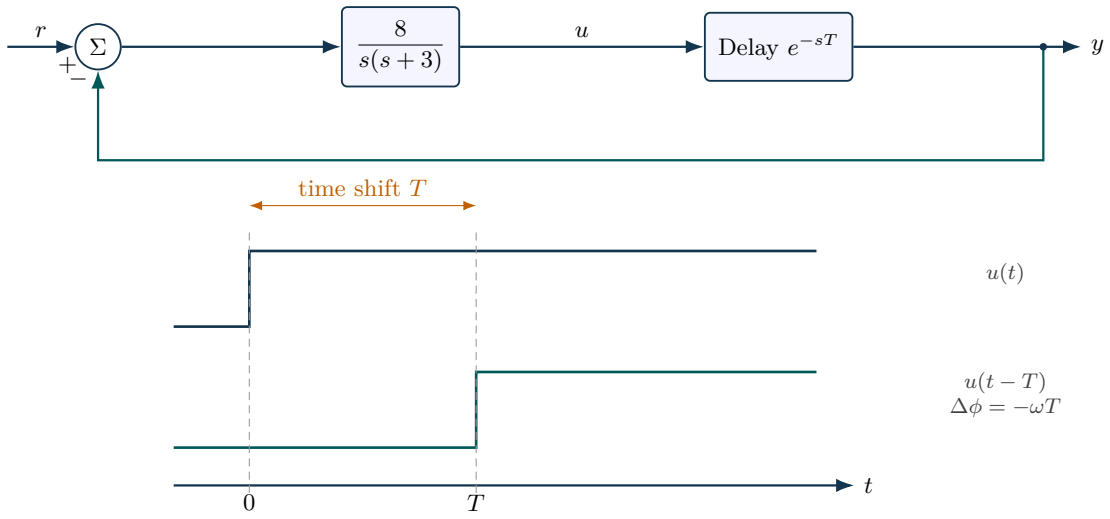
6 Design decision

Reserve phase for unmodeled sensor and actuator dynamics rather than spending the entire mathematical delay budget. Packet loss, variable delay and sampled-data effects are outside a constant-delay continuous-time approximation.

7 What to learn from the experiment

Exact delay phase supplies the stability-budget calculation. Pade models are checked approximations used to visualize nominal responses.

8 System sketch

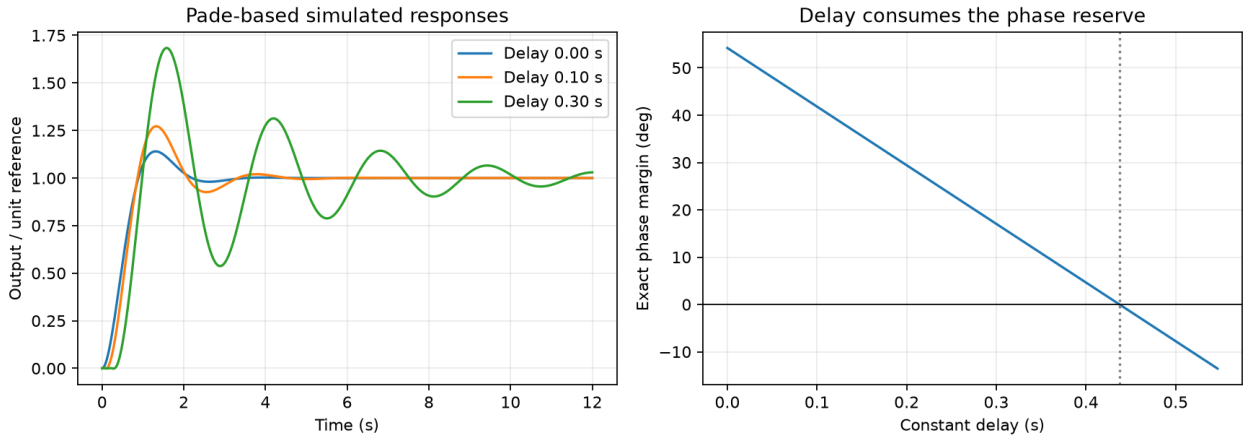


Equal-height unit steps, drawn on vertically offset tracks for clarity.

Figure 1. Constant pure delay shifts a signal in time and contributes frequency-dependent phase lag. The sketch shows ideal steps, not experimental records.

9 Executed computational evidence

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Delay 0.00 s: exact margin=54.207784 deg; Pade margin=54.207784 deg
Delay 0.10 s: exact margin=41.814427 deg; Pade margin=41.814427 deg
Delay 0.30 s: exact margin=17.027715 deg; Pade margin=17.027715 deg
Crossover=2.163049 rad/s; ideal delay budget=0.437394 s
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10 Method references

The equations use standard balance laws and feedback-control methods. Further explanation: https://apmonitor.com/pdc/ and https://fbsbook.org/. Implementation documentation: https://python-control.readthedocs.io/en/0.10.2/. See the repository reference catalog for individual topics and videos.